



A generic physics-informed neural network framework for lithium-ion batteries state of health estimation

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ABSTRACT

Accurate, practical, and robust battery state of health (SOH) assessment is crucial for the efficient and reliable operation of electric vehicles. However, reliable and precise battery SOH estimation remains challenging due to the complexity of battery aging models and the difficulty in parameter identification. In this paper, a general physics-informed neural network (PINN) framework is proposed, which uses a two-branch structure to achieve accurate estimation of SOH. Specifically, the degradation model is established by reconstructing the solid electrolyte interphase (SEI) film growth model using Universal Differential Equations (UDE). The extracted features are input into the first multilayer perception (MLP) branch to generate a preliminary SOH estimation. These features, initial SOH and the partial derivative of SOH to features are innovatively input into the degradation model represented by the second MLP branch, and the SEI growth model is explicitly simulated. Finally, the SEI film volume prediction results are mapped back to SOH. An applied voltage window (approximately 71 %–90 % state of charge range) is designed to extract 13 effective health indicators, making the method applicable to different battery types and charge-discharge protocols. Additionally, 83 batteries and 34,696 data samples including five experimental LFP/graphite batteries and three others with datasets are used to validate proposed approach. The mean absolute percentage error (MAPE) is 0.97 % across all test sets. Compared with other network architectures, the proposed PINN framework demonstrates excellent performance in all normal, small samples and transfer tests.

1. Introduction

With the intensification of global climate change and increasingly severe environmental pollution, the green transition in the transportation sector has become a critical pathway toward achieving carbon neutrality goals. As a key alternative to traditional internal combustion engine vehicles, electric vehicles (EVs) demonstrate significant potential in reducing carbon emissions and improving air quality[1–4]. In 2023, the global EV fleet surpassed 40 million units, marking a 35 % increase compared to 2022. It is projected to account for over 30 % of new vehicle sales by 2030. Amid this rapid development, lithium-ion batteries (LIBs) have emerged as the core component of EV powertrains due to their superior overall performance. However, as the market expands, increasingly stringent requirements are being placed on LIBs in terms of

cycle life, safety performance, and sustainability, driving both academia and industry to accelerate the development and application of next-generation battery technologies[5–9].

Accurate estimation of LIBs state of health (SOH) holds critical theoretical and practical significance for Battery Management Systems (BMS) and the optimization of energy storage systems[10–13]. As a key indicator characterizing battery degradation, precise SOH assessment not only directly impacts the reliability of Remaining Useful Life (RUL) predictions but also plays a vital role in optimizing system safety, cost-effectiveness, and energy efficiency management.

Currently, mainstream SOH estimation methods can be broadly categorized into two major approaches [14,15]: model-based methods and data-driven methods. Model-based methods require the reproduction of battery dynamic characteristics, such as Equivalent Circuit

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Models (ECMs) [16] and Electrochemical Models (EMs) [17]. These approaches aim to extract degradation-related features for battery SOH measurement. For instance, ECMs can capture internal ohmic resistance for SOH evaluation. However, such methods increase implementation complexity due to their demand for extensive preliminary experiments, parameter measurements, and substantial computational resources. EMs, which directly describe internal battery aging processes through electrochemical theory and physical equations, serve as crucial theoretical foundations for current battery health research. For example, Yang et al. [18] derived a lithium battery aging model by analyzing solid electrolyte interphase (SEI) growth. These methods possess clear physical interpretability and can deeply reveal battery degradation mechanisms, making them suitable for theoretical studies across different battery systems. Nevertheless, their high model complexity and reliance on precise parameter calibration may lead to insufficient real-time performance in practical applications.

In contrast, data-driven methods do not rely on explicit physical equations but instead employ machine learning algorithms to directly extract statistical relationships between SOH and characteristic parameters from battery operational data (e.g., voltage, current, temperature) [19–21]. Various deep learning and machine learning approaches have been developed for this purpose. For instance, Liu et al. [22] utilized a deep residual convolutional neural network to extract multi-dimensional features, followed by a fully connected network for SOH prediction. Liang et al. [23] developed a quantum convolutional neural network (QCNN) that enables precise, stable, and transferable state-of-health assessment while maintaining low computational complexity and minimal training data demands. While data-driven methods demonstrate superior accuracy and computational efficiency on specific datasets, their performance heavily depends on the quality of feature engineering. Current literature presents numerous feature extraction approaches tailored to particular datasets and charge/discharge conditions, yet insufficient attention has been given to feature generalizability [24]. Most of the existing features are extracted from the discharge curve, which are very sensitive to user-specific protocols [25]. In addition, the strong SOH-related features in battery chemistry (e.g., LiFePO_4) may exhibit weak correlations in other (e.g., NCM) due to differences in degradation mechanisms. This not only compromises model generalizability but also leads to substantial repetitive computational resource consumption. Therefore, this paper develops a strategy based on the charging curve, focusing on extracting key features from the short-period data in this interval, and analyzing the correlation between features and SOH [26]. Moreover, purely data-driven approaches lack physical constraints and may generate predictions that violate electrochemical principles under out-of-distribution scenarios, significantly compromising reliability in engineering applications. These limitations have motivated researchers to explore novel hybrid modeling approaches that combine the physical rigor of mechanistic models with the adaptability of data-driven methods [27].

In recent years, physics-informed neural network (PINN) has demonstrated remarkable advantages in the field of battery SOH estimation [28]. By integrating electrochemical mechanisms with deep learning, PINNs effectively address the aforementioned limitations. However, the existing PINN in SOH estimation faces the problems of complex loss function construction, high training cost and insufficient universality. Hofmann et al. [29] directly embedded the lithium-ion diffusion equations from the pseudo-2-dimensional (P2D) model into the loss function for SOH estimation. This approach exhibits strong physical consistency. However, it suffers from complex loss function construction and substantial training difficulties. Deng et al. [30] employed neural networks to estimate intermediate variables in the SEI growth process, subsequently deriving SOH through degradation models. Although this method reduces some equations and unknown parameters, its applicability remains limited. Jishnu et al. [31] proposed a degradation modeling framework based on Universal Differential Equations (UDE), which integrates complex aging models into neural

networks to learn partial degradation mechanisms. However, the framework of this method is hindered by the high training cost in the solution process. In the field of battery SOH estimation, current research faces a fundamental challenge in effectively integrating complex electrochemical reaction equations with neural networks. To address this issue, our study presents an innovative reconstruction of the SEI layer growth model into a more versatile UDE framework, enabling precise mathematical characterization of key factors influencing battery performance degradation. Building upon this foundation, we achieve a deep integration of electrochemical governing equations with deep learning architectures through high-fidelity neural network fitting of the established aging model.

Based on the above analysis, the contributions of this paper are as follows:

1. Different from the discharge process with individual dependence variability, the charging voltage behavior shows superior regularity and repeatability in the 71 %–90 % state of charge window. We focus on extracting key features from the short-period data in this interval and analyze the correlation between features and SOH. It is concluded that features can significantly enhance the generalization ability of the model.
2. Addressing the challenges of complex SEI growth modeling and parameter identification, we establish a degradation model of SEI growth based on UDE reconstruction, and effectively capture its dynamic evolution using neural networks.
3. Based on the established degradation model, a generic PINN structure is proposed for battery SOH estimation. The extracted features are input into the first multi-layer perception (MLP) branch to generate a preliminary SOH estimation. These features, initial SOH and the partial derivative of SOH to features are input into the established degradation model represented by the second MLP branch to dynamically capture the SEI growth process. Finally, it is mapped back to SOH.
4. To validate the effectiveness of the proposed method, we conducted systematic experimental verification. The datasets are constructed for variation including lifecycle data from five batteries of 142Ah experiment and 78 other batteries, which cover different chemical systems and charge-discharge protocols. The proposed method is systematically verified and compared with other methods. Finally, the effectiveness of the proposed transfer learning of PINN is verified by cross-dataset.

2. Article overview

This paper proposes a generic PINN framework flow for SOH estimation in LIBs, which is shown in Fig. 1. Including data generation module, feature extraction stage, PINN framework design and transfer learning.

In the data generation module (Fig. 1(a)), a new data set containing five battery life cycle data was constructed, and three public data sets were combined to contain a total of 83 batteries and 34696 test samples. Fig. 1(a) shows the basic information of the battery in four datasets and the degradation trajectories under different datasets.

In the feature extraction stage (Fig. 1(b)), we innovatively select short period data from the pre-full charge as the analysis target and extract statistically significant feature parameters as model inputs. This applied charging fragment is available across various battery datasets and effectively overcomes the adaptability limitations of traditional feature extraction methods. Consequently, our approach demonstrates excellent applicability across diverse battery chemistries and charge/discharge protocols.

The generic PINN framework constructed in this study adopts a two-branch structure design (Fig. 1(c)). One is the functional relationship between the battery characteristic parameters and SOH, $u = f(n, x)$, where, n is the number of cycles, x is the feature vector, and u is the SOH

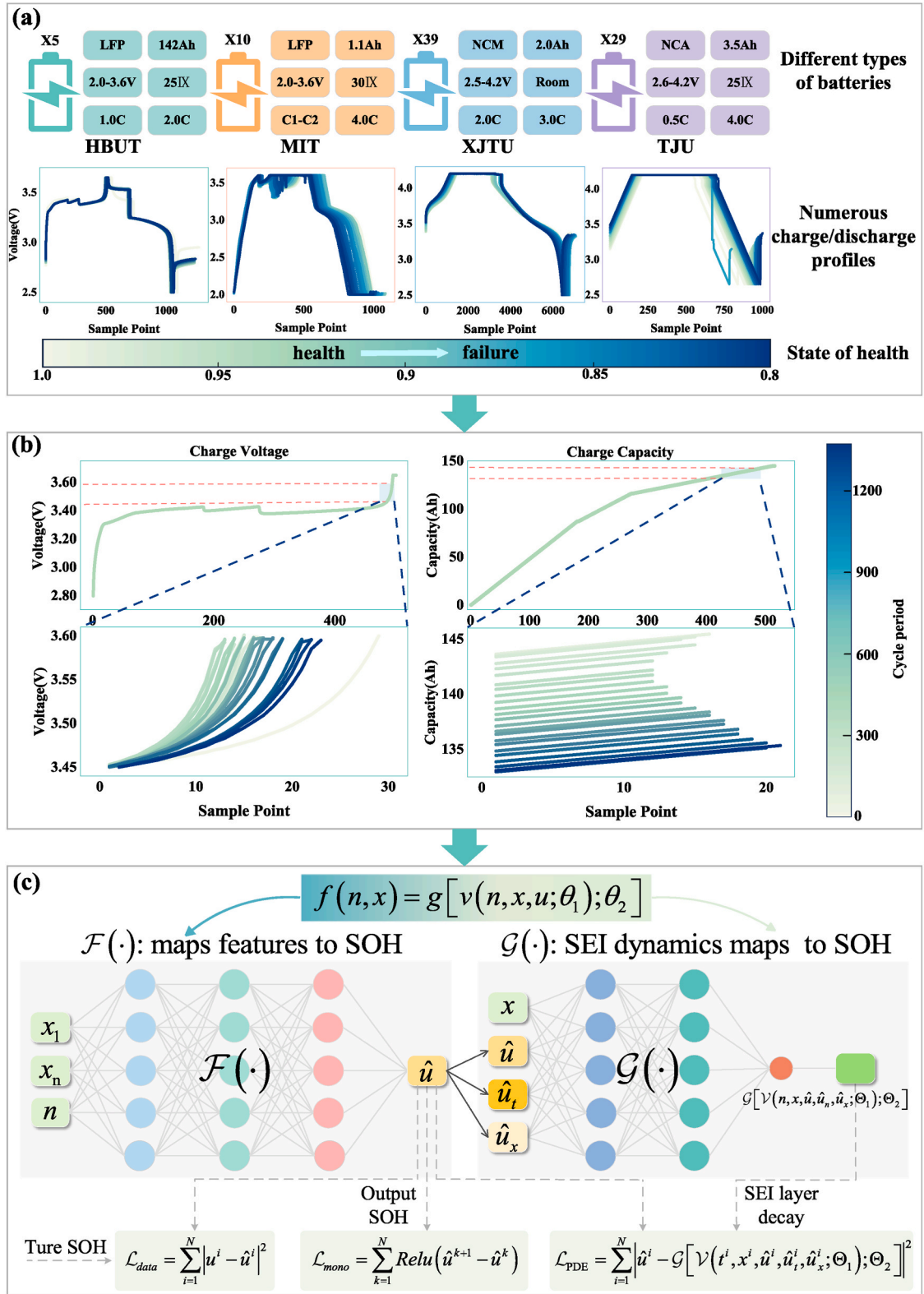


Fig. 1. shows the generic PINN SOH estimation method framework for lithium-ion batteries proposed in this study. (a) The differences in electrochemical characteristics of different cathode materials (including LFP, NCM and NCA) and the diversity of degradation trajectories caused by user-specific charging and discharging strategies; (b) The short-period data in the charge range of 71–90 % are innovatively selected as the feature extraction window. By analyzing the growth law of the voltage/capacity curve at the initial stage of the second cycle and the whole life cycle (sampling at intervals of 50 cycles), the characteristic parameters with time series characterization ability are extracted. (c) A dual-branch neural network structure is constructed, where $\mathcal{F}(\cdot)$ represents the mapping relationship between the battery characteristics and the SOH value, $\mathcal{G}(\cdot)$ represents the dynamic process of SEI film growth to battery degradation, and the physical mechanism and data-driven organic integration are realized by equation constraints (u_n and u_x represent the partial derivatives of SOH to period and characteristics, respectively), where u and $\hat{u}$ represent the real and predicted SOH values, respectively, and the superscript i denotes the index of individual samples in the dataset.

estimation value. The other one is the nonlinear functional relationship between the growth dynamics of SEI film and SOH which is derived by the UDE of the degradation modeling, $u = g[v(n, x, u; \theta_1); \theta_2]$, where, θ_1 , θ_2 are the parameterizations of the explicit equations. Considering that $f(\cdot)$ and $g(\cdot)$ in the actual battery system are affected by multi-factor coupling and are difficult to express explicitly, this study innovatively uses fully connected neural networks $\mathcal{F}(\cdot)$ and $\mathcal{G}(\cdot)$ to perform data-driven modeling on these two functions respectively. This hybrid modeling strategy not only retains the physical meaning of the electrochemical process but also achieves effective characterization of complex nonlinear relationships through neural networks. The training process of this study adopts a multi-objective loss function design, which contains three key optimization items. The data fitting term $\mathcal{L}_{data}$ minimizes the mean square error between the predicted SOH and the actual measured value to ensure the prediction accuracy of the model. The physical constraint term $\mathcal{L}_{mono}$ introduces monotonic regularization constraints to ensure that the model output conforms to the physical law of irreversible attenuation of battery capacity. The equation constraint term $\mathcal{L}_{PDE}$ incorporates the battery aging equation as a constraint.

In order to verify the generalization performance of the PINN proposed in this paper, we design a cross-cell type transfer learning experiment. Specifically, on the battery data sets with different chemical compositions, the SEI film growth model $\mathcal{G}(\cdot)$ parameters are frozen, and only the SOH mapping network $\mathcal{F}(\cdot)$ is fine-tuned. The experimental results show that the framework can effectively characterize the complex mapping relationship between the growth dynamics of SEI film and SOH. This result not only confirms the effectiveness of PINN in battery SOH estimation but also provides a new research idea for mechanism-based battery aging modeling.

3. Data sources

To validate the applicability of the proposed method, this study comprehensively incorporates aging data from batteries with different types and chemistries, comprising 34,696 test samples from 83 battery cells. The experimental data consists of four components: the HBUT dataset collected from our independently conducted battery cycling aging experiments; the MIT battery dataset publicly available from Severson et al. [32]; the XJTU battery dataset provided by Wang et al. [28]; and the TJU battery dataset released by Zhu et al. [33]. Table 1 summarizes the fundamental characteristics of all battery samples across the datasets.

The HBUT dataset includes cycling data from five lithium iron phosphate/graphite batteries tested under multi-stage charging protocols and a constant-current discharge protocol. These batteries were manufactured by Sunwoda, with a nominal capacity of 142 Ah and a nominal voltage of 3.3 V. The cells were cycled at 25 °C using multi-stage fast-charging strategies and a fixed discharge protocol until failure. The charge cutoff voltage was set to 3.65 V, and the discharge cutoff voltage was 2.5 V.

The MIT dataset comprises 10 commercial phosphate/graphite

batteries (model APR18650M1A, A123 Systems), with a standard capacity of 1100 mAh and a rated voltage of 3.3 V. All cells were tested under a constant ambient temperature of 30 °C, following a protocol that included 10 different multi-stage discharge strategies. The charging phase employed either single-step or two-step fast-charging methods, while the discharge phase was uniformly conducted at a 4 C-rate.

The XJTU dataset was constructed using 39 commercial NCM/graphite batteries (manufactured by LISHEN), with 2000 mAh nominal capacity and 3.6 V rated voltage. The operational voltage range was set between 2.5 V and 4.2 V. This dataset investigated the effects of four multi-stage discharge protocols, including fixed charge-discharge modes and periodic constant-current random discharge strategies. All batteries were cycled at room temperature until reaching the failure criteria.

The TJU dataset contains 9 NCA batteries with a nominal capacity of 3500 mAh and an operational voltage range of 2.65–4.2 V; 10 NCM batteries with a nominal capacity of 3500 mAh and an operational voltage range of 2.5–4.2 V; 10 NCM + NCA hybrid batteries with a nominal capacity of 2500 mAh and an operational voltage range of 2.5–4.2 V. All cells were cycled under different charging strategies until failure.

4. Feature extraction

Aiming at the characteristics of complex and changeable working conditions in the actual operation of electric vehicles, we innovatively propose a multi-feature extraction framework specifically designed for charging process analysis. This method systematically constructs a set of health indicators (HIs) that can comprehensively characterize the SOH of the battery by deeply mining the effective information in the charging curve, which lays a data foundation for the subsequent accurate estimation of the battery state.

Based on the analysis of the characteristics of the battery charging curve, this study selects a specific voltage interval for HI extraction. A feature extraction window of 3900–4100 mV and 3400–3550 mV is set for the ternary lithium battery and the lithium iron phosphate battery (corresponding to about 71–90 % state of charge interval), which covers the most representative voltage change area in the constant current charging stage [22], as shown in Fig. 1 (b). By extracting the voltage time series data of this interval as the first kind of HI sequence, the key information reflecting the aging state of the battery can be effectively obtained.

Based on the analysis of the aging mechanism of lithium-ion batteries, it is found that the battery decay will significantly affect its charging characteristics. In the same charging voltage range, the voltage rise rate of the aging battery is accelerated and the charging capacity is reduced [32]. This phenomenon is due to the combined effect of lithium-ion loss and active material decay, which is manifested in the peak shift of the incremental capacity (IC) curve and the reduction of the integral area [22]. The integrated area under IC curves serves as a direct indicator of remaining capacity. We quantitatively assess battery degradation severity through systematic extraction of charge capacity

Table 1
Basic conditions of each dataset.

Dataset	Batch	Battery type	Nominal capacity(mAh)	Cut-off voltage(V)	Temperature (°C)	Total number	Number of test set
HBUT	–	LiFePO ₄	142000	2.0–3.65	25	5	1
MIT	–	LiFePO ₄	1100	2.0–3.6	30	10	2
XJTU	1	LiNi _{0.5} Co _{0.2}	2000	2.5–4.2	Room temperature	8	2
	2	Mn _{0.3} O ₂				15	3
	3					8	2
	4					8	2
TJU	1	Li _{0.86} Ni _{0.86}	3500	2.65–4.2	25,35,45	10	2
		Co _{0.11} Al _{0.03} O ₂					
	2	Li _{0.84} Ni _{0.83}	3500	2.5–4.2	25,35,45	10	2
		Co _{0.11} Al _{0.07} O ₂					
	3	LiNiCoMnO ₂ and LiNiCoAlO ₂	2500	2.5–4.2	25	9	2

sequences within defined voltage windows. This method constructs a standardized feature sequence with clear physical meaning by establishing a voltage-capacity function relationship, which provides a reliable basis for SOH evaluation.

With the aging of the battery, the increase of electrode polarization leads to the acceleration of the charging rate in the same voltage range. Therefore, under the condition of constant current charging, the charging time required for the battery to complete a specific voltage range (such as 3.9–4.1 V) is significantly shortened. In this study, the charging time in a fixed voltage range is proposed as the third type of HI.

The mean, standard deviation, kurtosis, skewness, curve slope and curve entropy were extracted from the first-class HI sequence and the second-class HI sequence [28]. These features were numbered from 1 to 12, and the third-class HI number was 13. Table 2 gives 13 health index points and calculation methods. Fig. 2(a) shows the visualization results of the correlation between health features and SOH extracted from four datasets. In order to quantify the validity of the features, the linear correlation between the features of each data set and SOH is evaluated by calculating the Pearson correlation coefficient. The specific statistical results are shown in Fig. 2(b).

The experimental results show that the correlation between the characteristics of different battery chemical compositions and SOH is significantly different, and the correlation between the characteristics of the same battery chemical composition and SOH is similar. As shown in Fig. 2(b), although the HBUT and MIT datasets (both LiFePO₄ batteries) have significant differences in capacity and charge-discharge strategies, the extracted features show similar correlations with SOH (such as strong positive correlation in feature 7 and strong negative correlation in feature 12). This rule was further verified in the XJTU data set (NCM battery), and the correlation between its features and SOH remained consistent (such as the strong negative correlation of features 3–4, 8, 12–13, and the strong positive correlation of features 1–2, 5–7), indicating that the chemical composition is the decisive factor affecting the effectiveness of features. From the datasets of different chemical systems, most of the features are different from the sensitivity of SOH. For example, 1–6, 8, 13 in XJTU showed strong correlation, while in HBUT showed weak correlation. 11 in HBUT showed strong correlation, while XJTU showed weak correlation. In particular, feature 10 shows a

weak correlation in HBUT, MIT and XJTU, while a strong correlation in TJU. It is the different sensitivity of each feature to SOH that proves that the extracted features are suitable for different chemical systems, which also plays a key role in the subsequent transfer learning between different datasets.

V_{HI} charging voltage sequence, Q_{HI} charging capacity sequence, t_{end} end time of charging sequence, t_{start} start time of charging sequence.

5. Methods

5.1. Battery degradation model

The primary characteristics of battery aging manifest as continuous capacity degradation, increasing internal resistance, and evolving voltage characteristics, typically exhibiting a descending trend. For precise characterization of battery capacity fade dynamics, researchers investigated the mechanisms of lithium-ion battery performance deterioration over time and under various operating conditions through mechanistic modeling. Through mechanistic analysis of stress-induced particle fracture and SEI formation dynamics, Li et al. [34] quantitatively correlated the capacity degradation rate with the volumetric expansion characteristics of the SEI layer.

$$\frac{dQ_N}{dN} = -\frac{n_{SEI}\rho_{SEI}F}{M_{SEI}} \frac{dV_{SEI}}{dN} \quad (1)$$

where, Q_N represents the capacity of the cycle N times, n_{SEI} represents the Consumed lithium ion for 1 mol SEI layer formation, ρ_{SEI} represents the Density of SEI films, M_{SEI} represents the Molecular weight of compounds SEI, F is the Faraday constant, V_{SEI} represents the SEI film volume, including the formation of the initial SEI film and the SEI film grown on the pyrolysis surface.

Unfortunately, the difficulty in modeling the aging model and the difficulty in identifying the aging parameters bring many difficulties to the study of battery aging. In response to the problems, researchers have developed multiple empirical modeling approaches to characterize battery capacity degradation as a function of temporal evolution or cycling conditions, including linear models [35], power-law models [36] and fault prediction models (FFM) [37].

The existing empirical model simplifies battery degradation into a time univariate function, which fails to fully reflect the complexity of the actual aging mechanism. Studies have shown that battery aging is the result of a combination of factors including storage time, temperature, charge and discharge rate, etc. Among them, the growth of SEI film is the key degradation mechanism, and its evolution process has a coupling relationship with the above multi-physical field factors. The electrochemical-thermal-mechanical coupling model proposed by Luo et al. [38] more accurately describes the actual degradation behavior of the battery by establishing the multivariate function relationship between the aging rate and voltage, current, temperature and stress.

Therefore, the battery SOH is described as a multivariate function of cycle times, charging voltage, charging capacity, and charging time.

$$u = f(n, x) \quad (2)$$

Where, n is the number of cycles, x is the characteristic parameter, u represents SOH. In view of the inherent complexity of SEI formation modeling, we use the UDE framework to re-express the SEI growth kinetics as an explicit parametric representation. Eq. (1) reveals that there is a linear relationship between battery capacity and SEI film volume. Without loss of generality, the SEI film volume can be described as:

$$V_{SEI} = v(n, x, u; \theta) \quad (3)$$

The above equation represents an explicit formulation parameterized by θ , where $v(\cdot)$ denotes function of n , x , and u . The $v(\cdot)$ describes the nonlinear kinetic behavior of SEI layer growth within the battery. Through appropriately modifying this nonlinear function, diverse

Table 2
Calculation method of health index points.

Index	HI points	Illustration	Calculation
1	V_{mean}	Average value of V_{HI}	$V_{mean} = \frac{1}{n} \sum_{i=1}^n V_{HI}(i)$
2	V_{std}	Standard deviation of V_{HI}	$V_{std} = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (V_{HI}(i) - V_{mean})^2}$
3	V_{kur}	Kurtosis of V_{HI}	$V_{kur} = \frac{\sum_{i=1}^n (V_{HI}(i) - V_{mean})^4}{(n-1)V_{std}^4}$
4	V_{ske}	Skewness value of V_{HI}	$V_{ske} = \frac{\sum_{i=1}^n (V_{HI}(i) - V_{mean})^3}{(n-1)V_{std}^3}$
5	V_{slope}	Curve slope of V_{HI}	$V_{slope} = \frac{V_{end} - V_{start}}{t_{end} - t_{start}}$
6	V_{ent}	Curve entropy of V_{HI}	$V_{ent} = -\sum_{i=1}^n V_{HI}(i) \cdot \log(V_{HI})$
7	Q_{mean}	Average value of Q_{HI}	$Q_{mean} = \frac{1}{n} \sum_{i=1}^n Q_{HI}(i)$
8	Q_{std}	Standard deviation of Q_{HI}	$Q_{std} = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (Q_{HI}(i) - Q_{mean})^2}$
9	Q_{kur}	Kurtosis of Q_{HI}	$Q_{kur} = \frac{\sum_{i=1}^n (Q_{HI}(i) - Q_{mean})^4}{(n-1)Q_{std}^4}$
10	Q_{ske}	Skewness value of Q_{HI}	$Q_{ske} = \frac{\sum_{i=1}^n (Q_{HI}(i) - Q_{mean})^3}{(n-1)Q_{std}^3}$
11	Q_{slope}	Curve slope of Q_{HI}	$Q_{slope} = \frac{Q_{end} - Q_{start}}{t_{end} - t_{start}}$
12	Q_{ent}	Curve entropy of Q_{HI}	$Q_{ent} = -\sum_{i=1}^n Q_{HI}(i) \cdot \log(Q_{HI})$
13	Δt	Charging time	$\Delta t = t_{end} - t_{start}$

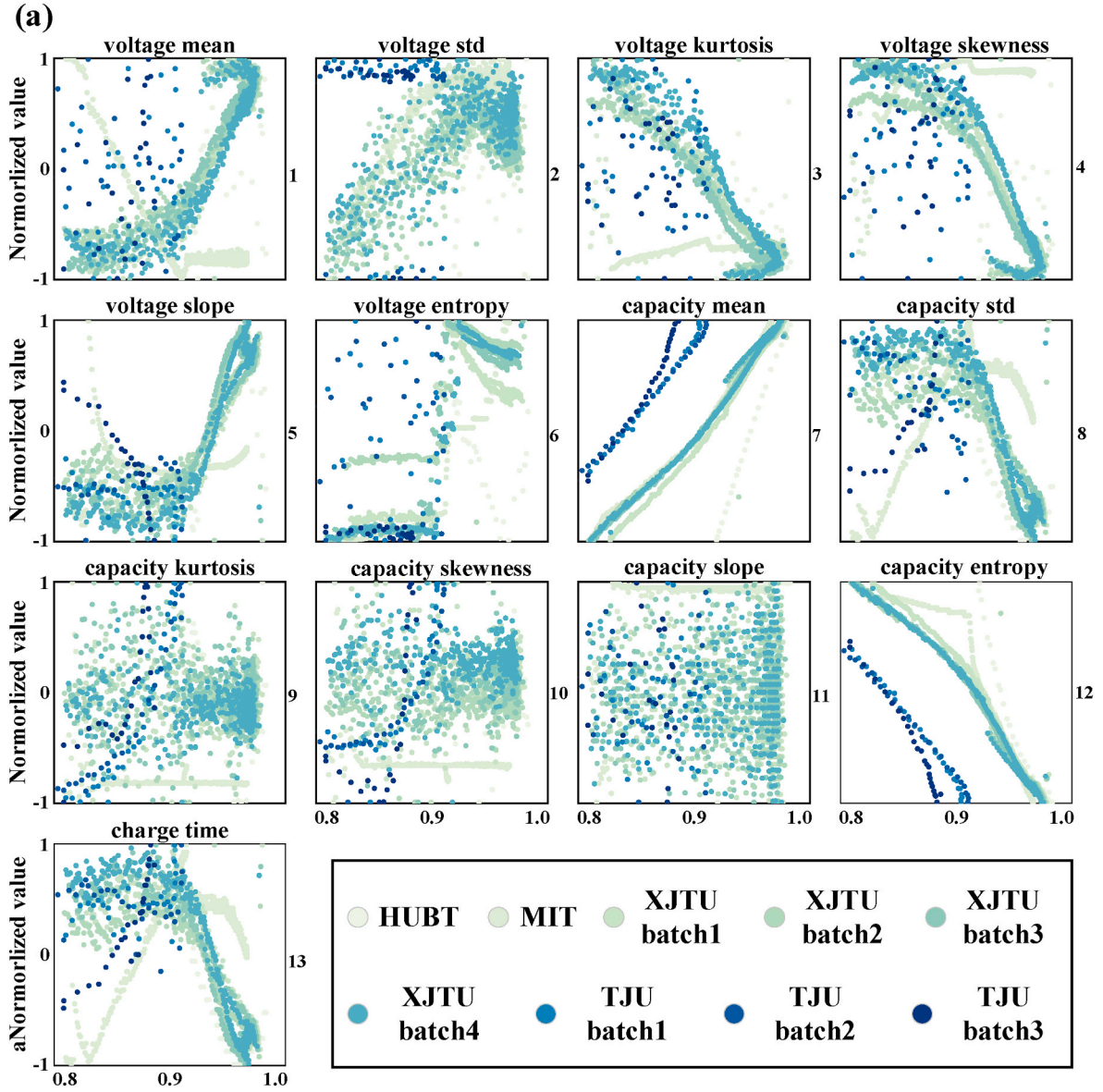


Fig. 2. shows the results of feature extraction and correlation analysis. (a) The feature distribution of the batteries in the four data sets. Each subgraph takes SOH as the abscissa and the normalized eigenvalue as the ordinate, and the right number corresponds to 13 feature sequences. The heat map of (b) quantifies the correlation strength between each feature and SOH, and the feature number is strictly corresponding to (a).

growth mechanisms can be systematically modeled.

5.2. SOH estimation based on PINN

A fundamental challenge persists in determining the exact mathematical expression for $v(\cdot)$. To address such issues, Kuzhiyil et al. [31] introduced a novel approach combining neural networks with physics-based differential equations through UDEs, facilitating the learning of partially characterized degradation processes. Raissi et al. [39] developed deep hidden physics models to effectively parameterize the unknown function. Drawing inspiration from Refs. [31,39], we adopt a neural network $\mathcal{V}(\cdot)$ with learnable parameters Θ_1 to approximate $v(\cdot)$, enabling data-driven discovery of SEI growth dynamics from given inputs n, x and associated partial derivatives. Consequently, Eq. (3) transforms into:

$$V_{SEI} \approx \mathcal{V}(n, x, u, u_n, u_x; \Theta_1) \quad (4)$$

In the formula, $u_n = \frac{\partial u}{\partial n}$, the neural network $\mathcal{F}(n, x; \phi)$ is used to model $f(n, x)$ by using the learnable parameter ϕ , and the automatic differential mechanism is used to calculate $u_n, u_x = \left[\frac{\partial u}{\partial x_1}, \frac{\partial u}{\partial x_2}, \dots \right]^T$ denotes the first-order partial derivative of u with respect to x . This approach offers significant advantages by effectively merging the approximation capabilities of machine learning with the physical interpretability of mechanistic models. Neural network $\mathcal{V}(\cdot)$ proposes a more elastic functional relationship for n, x, u and partial derivatives of various orders. Due to the approximate linear decay of capacity decay, as shown in Fig. 1(b), battery aging is dominated by first-order kinetics [31]. Secondly, the higher-order terms will significantly increase the training complexity of PINN. Therefore, our implementation only considers first-order derivative effects while neglecting higher-order derivatives. Then the neural network $\mathcal{G}(\cdot)$ with learnable parameter Θ_2 is used to map the obtained SEI film volume to the battery capacity.

In this paper, we define a physics-informed neural network $\mathcal{N}$ for the

(b)

HUBT	0.16	0.13	0.26	0.34	0.37	-0.25	0.99	-0.18	-0.44	0.01	0.97	-0.97	-0.28
	1	2	3	4	5	6	7	8	9	10	11	12	13
MIT	-0.83	0.82	-0.20	0.58	0.30	0.92	0.98	0.41	-0.28	-0.35	0.49	-0.94	0.36
	1	2	3	4	5	6	7	8	9	10	11	12	13
XJTU	0.94	0.57	-0.97	-0.94	0.84	0.74	0.99	-0.85	-0.10	0.01	-0.07	-0.98	-0.84
	0.91	0.74	-0.96	-0.92	0.83	0.89	0.99	-0.83	-0.02	0.01	0.04	-0.98	-0.83
	0.95	0.71	-0.98	-0.96	0.87	0.87	0.99	-0.87	-0.07	0.04	0.01	-0.99	-0.87
	0.88	0.76	-0.95	-0.90	0.87	0.90	0.99	-0.88	-0.11	-0.06	0.07	-0.98	-0.87
	1	2	3	4	5	6	7	8	9	10	11	12	13
TJU	0.24	0.03	-0.01	-0.03	-0.51	-0.16	0.99	-0.31	0.93	0.83	0.10	-0.99	0.44
	-0.15	-0.16	-0.37	0.48	-0.62	0.01	0.99	0.02	0.88	0.45	-0.41	-0.99	0.56
	0.18	-0.24	-0.18	0.26	-0.95	0.16	0.98	0.89	0.75	0.36	-0.03	-0.98	0.93
	1	2	3	4	5	6	7	8	9	10	11	12	13

Feature Number

Fig. 2. (continued).

study of battery capacity fading and SEI film growth:

$$\mathcal{H} := \mathcal{F}(n, x; \phi) - \mathcal{G}[\mathcal{V}(n, x, u, u_n, u_x; \Theta_1); \Theta_2] \quad (5)$$

Fig. 1 (c) shows the PINN structure.

The calculation formula of battery SOH is expressed as:

$$u^k = f(k, x) = \frac{Q^k}{Q^0} \quad (6)$$

where, Q^k denotes the capacity at cycle number k , while Q^0 corresponds to the nominal capacity. When $n = k$, the SOH value u^k becomes equivalent to the degradation model $f(\cdot)$. At the sample point i , make $\mathcal{H}(n^i, x^i) = 0$, approximate the obtained Eq. (1). Therefore, the PINN optimization process must satisfy the equation constraints defined in Eq. (1):

$$\mathcal{L}_{PDE} = \sum_{i=1}^N |\mathcal{H}(n^i, x^i)|^2 \quad (7)$$

where, i denotes the i -th data instance within a dataset of cardinality N . The composite loss function comprises two key components including data fidelity loss and monotonicity constraint loss:

$$\mathcal{L}_{data} = \sum_{i=1}^N |u^i - \hat{u}^i|^2 \quad (8)$$

$$\mathcal{L}_{mono} = \sum_{k=1}^M Relu(\hat{u}^{k+1} - \hat{u}^k) \quad (9)$$

Where, $\hat{u}$ devotes the estimated SOH, while M indicates the total cycle count. $Relu(\cdot)$ is the modified linear unit, and the monotonicity constraint $\mathcal{L}_{mono}$ is that SOH should conform to the monotonically decreasing nature of the battery degradation process. The total loss function is expressed as:

$$\mathcal{L} = \mathcal{L}_{data} + \alpha \mathcal{L}_{mono} + \beta \mathcal{L}_{PDE} \quad (10)$$

where, α and β are trade-off parameters. Adam optimizer is used to jointly train α , β and network weights. The hardware configuration is 13th Gen Intel (R) Core (TM) i9-13980HX, and the framework version is PyTorch 1.12.1 + CUDA 11.6. In order to ensure the fairness of the different method comparison, the number of network layers is 10, the number of neurons per layer is 50, the number of iterations is 5000, and the learning rate is set to 0.001.

5.3. Fine-tuning based on PINN for transfer learning

Current research on transfer learning for SOH estimation has primarily focused on adapting models across varying charge-discharge protocols, with limited investigation into different battery material systems. Since the growth of SEI film mainly depends on the negative

electrode [40], while the negative electrode of lithium iron phosphate and ternary lithium battery is graphite, and there is not much difference. The only difference is that the metal of the positive electrode is different, and the transition metal migrates to the negative electrode to accelerate the growth of the SEI film. Therefore, due to the generalization of SEI growth kinetics between material specificity, this paper innovatively explores the migration ability between battery datasets of different chemical systems. Transfer learning uses fine-tuning techniques to pre-train the model by using massive source domain data for parameter optimization and then fine-tunes on a small number of target domain datasets. In the experimental design, by combining four data sets in pairs, the adaptive performance of the model in cross-chemical composition scenarios was systematically evaluated.

This study proposes a transfer learning strategy based on physical mechanism. Theoretical analysis shows that the SEI film growth dynamic $\mathcal{G}(\cdot)$ has the universal characteristics of cross data sets, and the solution $\mathcal{F}(\cdot)$ is related to the specific working conditions. As shown in Fig. 3, the model training adopts a hierarchical optimization strategy and uses the source domain data pre-training to obtain the general $\mathcal{G}(\cdot)$ parameters. The $\mathcal{G}(\cdot)$ weight is frozen during migration, and only the $\mathcal{F}(\cdot)$ parameter is fine-tuning. Fine-tuning refers to train $\mathcal{F}(\cdot)$ with small sample data of the first battery in the target sets. Non-tuning refers to that the trained PINN $\mathcal{G}(\cdot)$ of the original sets is tested directly on the target set.

6. Validation

The PINN model constructed in this study uses 13 characteristic parameters and cycle times as input variables to predict SOH. In order to optimize the performance of the model, the min-max normalization method is used to uniformly scale all the feature parameters to the $[-1, 1]$ interval to eliminate the influence of dimensional differences on model training.

To demonstrate the advancement of the proposed PINN, we employ two comparative approaches: the PINN framework developed by Wang et al. [28] (hereafter referred to as XJT), and a conventional multilayer perceptron (MLP) with identical architecture and parameter count. The experimental dataset is partitioned into training, validation, and test sets at a ratio of 6:2:2, with test samples uniformly distributed across all datasets to ensure unbiased evaluation. The detailed sample distribution is presented in Table 1. In order to evaluate the robustness of the model, 10 times of repeated training and testing were performed in each experiment.

Fig. 4 (a) shows the SOH estimation effect of the model on four data sets. Under the HBUT dataset, the estimated SOH value is smaller than the true SOH value, but the absolute error is kept within 0.04. For the MIT data set, the SOH estimation value is in good agreement with the true value, and the absolute error is kept within 0.03. In the XJTU data set, the SOH distribution estimated by PINN is roughly the same under

four different batches, and the absolute error is kept within 0.05, which also shows that PINN can still maintain stable and reliable performance under different working conditions and individual differences of the battery. Similarly, the estimation error of SOH under the TJU dataset is also kept within 0.04.

Fig. 4 (b) is the comparison of our proposed PINN method with XJT and MLP. It can be seen from Fig. 4 (b) that under different data sets, the mean error of PINN is maintained at 2 %, the mean error of XJT in SOH estimation is maintained at 2.5 %, and the mean error of SOH estimated by the traditional MLP method is within 5 %. The accuracy of the proposed PINN is much higher than that of MLP. The accuracy of PINN is only 0.3 % higher than that of XJT, but the overall stability is greatly improved.

Table 3 compares the MAPE of the three models on four datasets. The experimental results show that the PINN method proposed in this study shows the best estimation accuracy in most cases, and the average MAPE is 0.74 % (HBUT), 0.84 % (MIT), 1.09 % (XJTU) and 0.95 % (TJU), respectively. In contrast, the corresponding error of the PINN framework of XJT is 0.79 %–1.13 %, while the error range of the MLP method is 1.16 %–3.51 %. In particular, it should be noted that the three models are completely consistent in terms of parameter quantity and structure, which further verifies that the superiority of the proposed method stems from its innovative modeling mechanism rather than parameter scale.

In order to verify the superior generalization performance of the PINN framework proposed in this paper compared with the pure data-driven method under the condition of limited training data, we designed a systematic small sample experiment. Four representative data sets of HBUT, MIT, XJTU batch1 and TJU batch1 were selected in the experiment. The number of training batteries (from single battery to four batteries) was gradually increased in each data set to simulate the data scarcity scenario in practical applications. The battery of the test set was the same as the battery in Table 1. The PINN model is compared with the MLP on the same training set and test set to ensure the fairness of the evaluation. The experimental results are shown on Fig. 5.

Fig. 5 shows that our proposed PINN achieves the best results in all datasets. Even using a small sample dataset of one battery to train the model, the MAPE of PINN on the four datasets does not exceed 2.8 %, 3.0 %, 3.8 % and 2.4 %, respectively, while the MAPE of MLP on the four datasets reaches 5.3 %, 5.2 %, 6.1 % and 4.3 %, respectively. The results show that the proposed PINN model can still maintain excellent performance under small sample conditions, which fully proves that the model has high prediction accuracy and stability under limited data. As the number of training samples increases, the performance of both models shows an upward trend, which is consistent with the general understanding in the field of machine learning: increasing the amount of training data helps to improve model performance. In particular, it should be pointed out that the PINN model trained with single battery data can achieve the performance level that the traditional MLP model requires 3–4 battery training data. This result highlights the significant

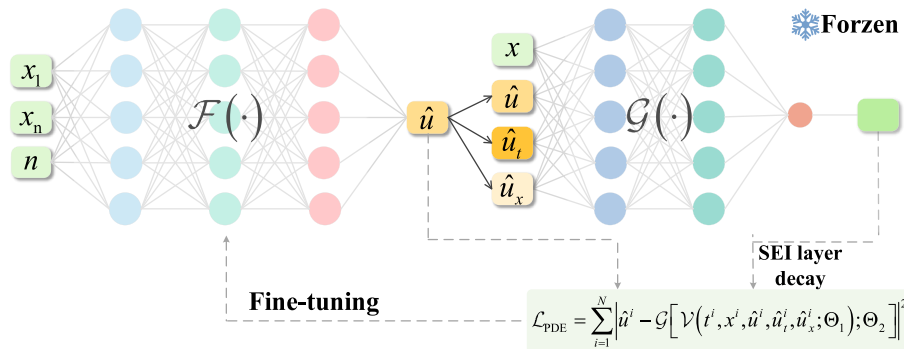


Fig. 3. PINN network parameter fine-tuning. PINN is trained on the source domain to obtain $\mathcal{F}(\cdot)$, $\mathcal{G}(\cdot)$ parameters. When testing PINN on the target domain, $\mathcal{G}(\cdot)$ is frozen, and the loss function is obtained to fine-tune the parameters in $\mathcal{F}(\cdot)$.

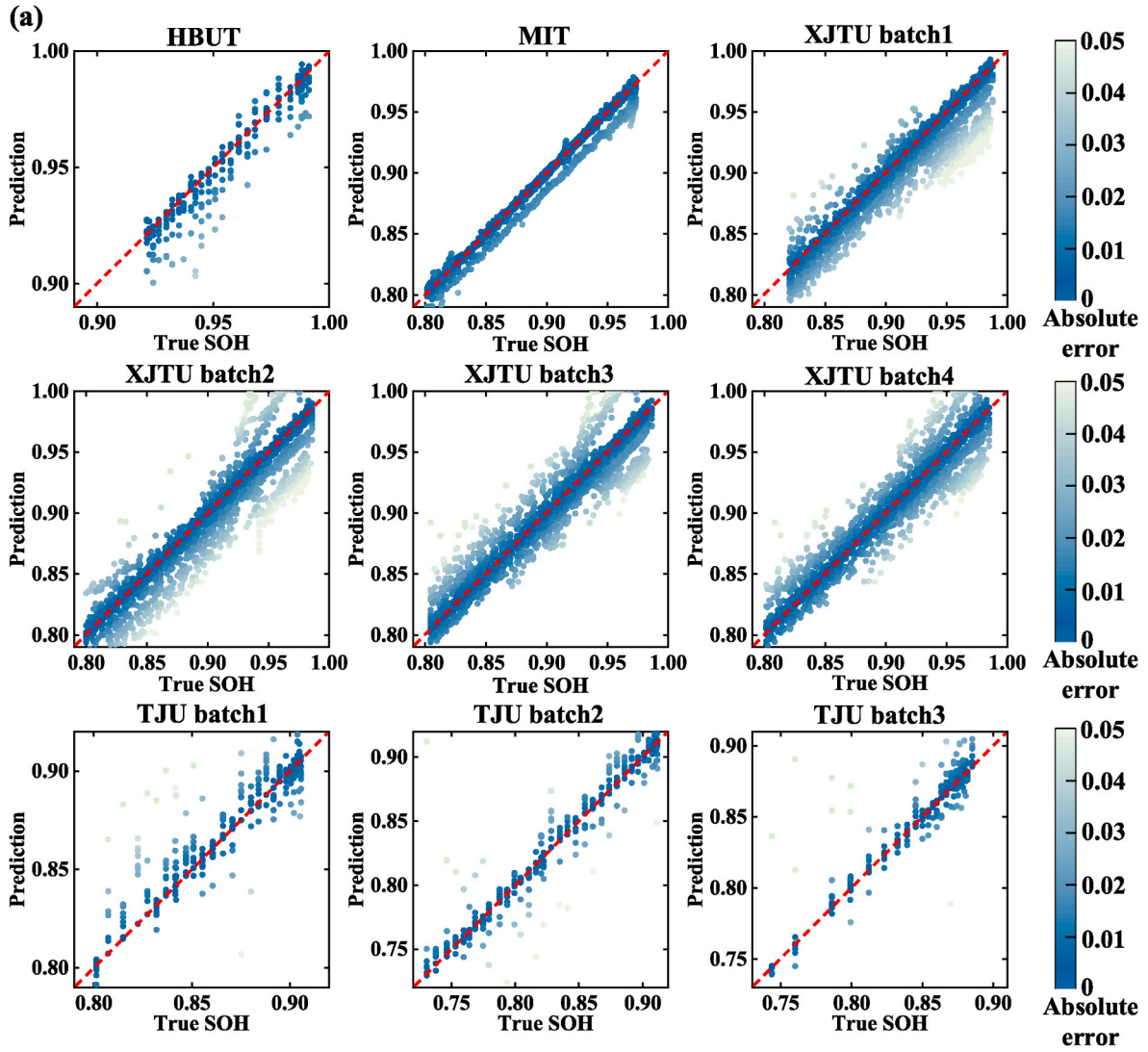


Fig. 4. presents the comparative performance analysis of SOH estimation. (a) The SOH estimation results of the proposed PINN on four datasets. Both predicted and real SOH are distributed near the diagonal, indicating that the model performs well. (b) The mean absolute error (MAE), MAPE and root mean square error (RMSE) distributions of the three models PINN, XJT and MLP on four datasets. Each error bar represents 10 independent experimental trials, with the central marker indicating the mean value.

advantages of the PINN method in small sample learning scenarios.

This study verifies the effectiveness of transfer learning through comparative experiments. The results are shown in Table 4. View the table by row, the first row represents the HBUT data set as the source domain, other data sets as the target domain, and so on. For XJTU and TJU data sets, we only use the data in XJTU batch 1 and TJU batch 1 to represent. The comparison results of the two are better expressed in bold, and all values are taken as the average of ten experiments.

The results show that the fine-tuning model achieves superior performance compared to its non-tuning model, exhibiting an approximately 25 % reduction in MAPE. Notably, the fine-tuning performance between the data of the same battery chemical composition is better than that of different battery chemical compositions. For example, HBUT is used as the source domain, and MAPE is 0.0117, 0.0240 and 0.0369 under the other three datasets, respectively. The results of MIT are better than XJTU and TJU, which may be because both HBUT and MIT are lithium iron phosphate batteries, making it easier to fine-tune network parameters. Similarly, with ternary lithium as the source domain, the effect of fine-tuning between the same ternary lithium is better than that of lithium iron phosphate. These findings strongly

confirm that the kinetic parameter $\mathcal{E}(\cdot)$ can extract the essential characteristics of generalization ability from the source domain data.

7. Conclusions

In this paper, we propose a SOH estimation method for LIBs based on a generic PINN framework. Aiming at the problems of strong feature dependence and insufficient model generalization ability of existing methods, we innovatively propose a general feature extraction method, which extracts voltage and capacity data from short-period data of battery charging state about 71–90 % and performs statistical features. And the correlation between features and SOH is analyzed. The experimental results show that the correlation between the characteristics of different battery chemical compositions and SOH is significantly different, and the correlation between the characteristics of the same battery chemical composition and SOH is similar. Future research can consider feature extraction methods that are equally applicable in specific scenarios.

In addition, a PINN containing the battery aging model is constructed. The extracted features are input into the first MLP branch to

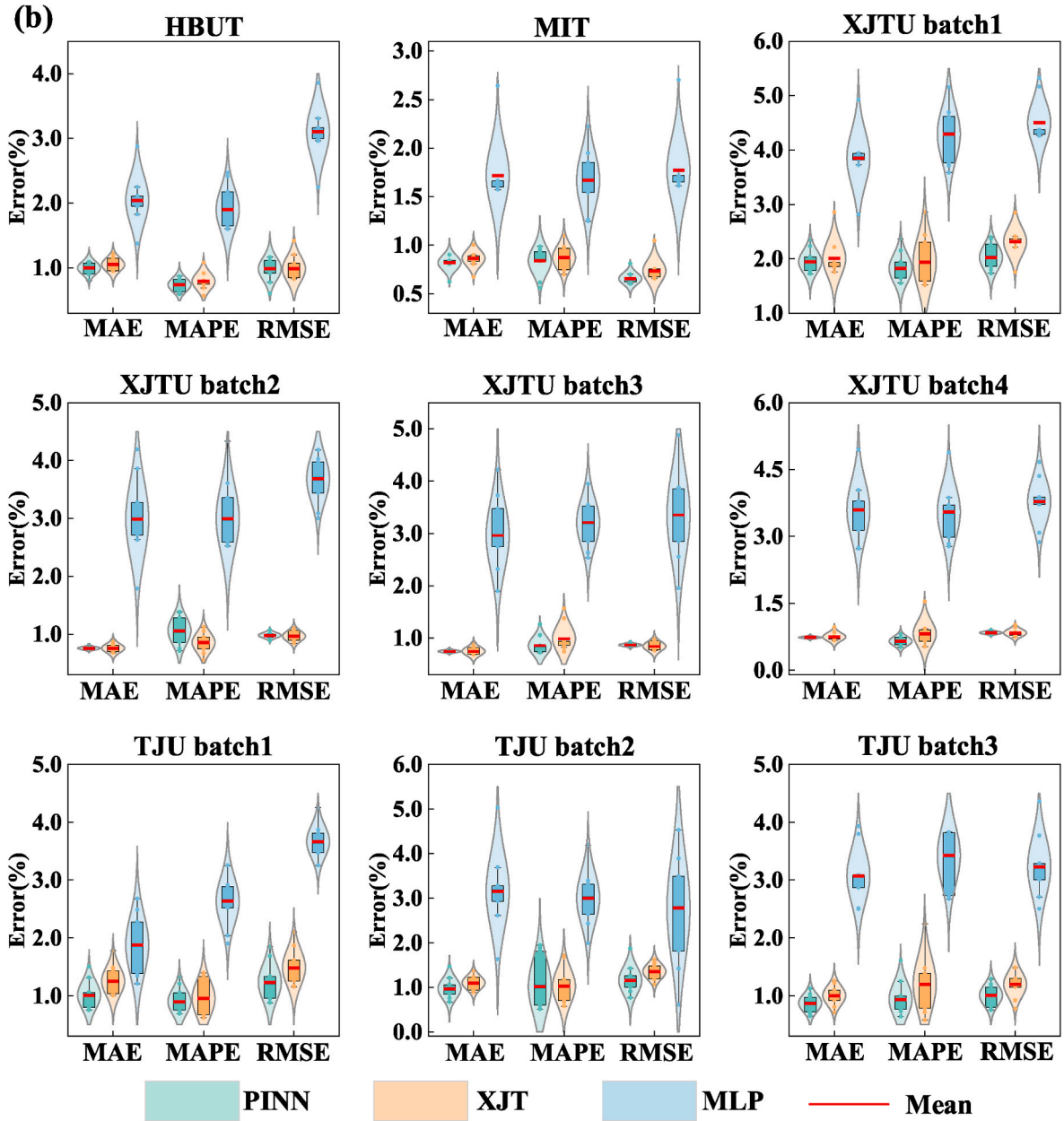


Fig. 4. (continued).

Table 3

The results of PINN with XJT and MLP on four datasets. The bold font is the best value. All values are taken as the average of ten experiments. All values are expressed in percentage.

Dataset	Batch	PINN		XJT		MLP	
		MAPE	RMSE	MAPE	RMSE	MAPE	RMSE
HBUT		0.74	0.98	0.79	0.98	1.90	3.10
MIT		0.84	0.65	0.87	0.73	1.16	1.77
XJTU	1	1.82	2.02	1.93	2.31	4.29	4.50
	2	1.06	0.97	0.85	0.96	2.99	3.68
	3	0.85	0.86	0.98	0.84	3.21	3.35
	4	0.65	0.83	0.81	0.83	3.55	3.78
TJU	1	0.90	1.23	0.95	1.48	2.63	3.66
	2	1.01	1.15	1.02	1.35	3.00	2.77
	3	0.94	1.00	1.20	1.19	3.42	3.23

generate a preliminary SOH estimate. These features, preliminary SOH, and partial derivatives of SOH to features are input into the established degradation model represented by the second MLP branch. Finally, the SEI film volume prediction results are mapped back to SOH. In this paper, the effectiveness of the proposed method is verified by system experiments. The method is fully verified on five self-built battery aging data sets and three other public data sets containing 78 batteries. The MAPE of the proposed PINN under different data sets is 0.97 %, which is greatly improved in accuracy and stability compared with the TJU and MLP methods. In order to verify the superior generalization performance of the PINN framework proposed in this paper compared with the pure data-driven method under the condition of limited training data, we designed systematic small sample experiments and verified the significant advantages of the PINN method in small sample learning scenarios.

In order to further evaluate the advantages of the method, a cross-domain migration experiment is designed. The fine-tuning technology in transfer learning is used to quickly adjust the model parameters by using a small amount of data in the target domain after pre-training the

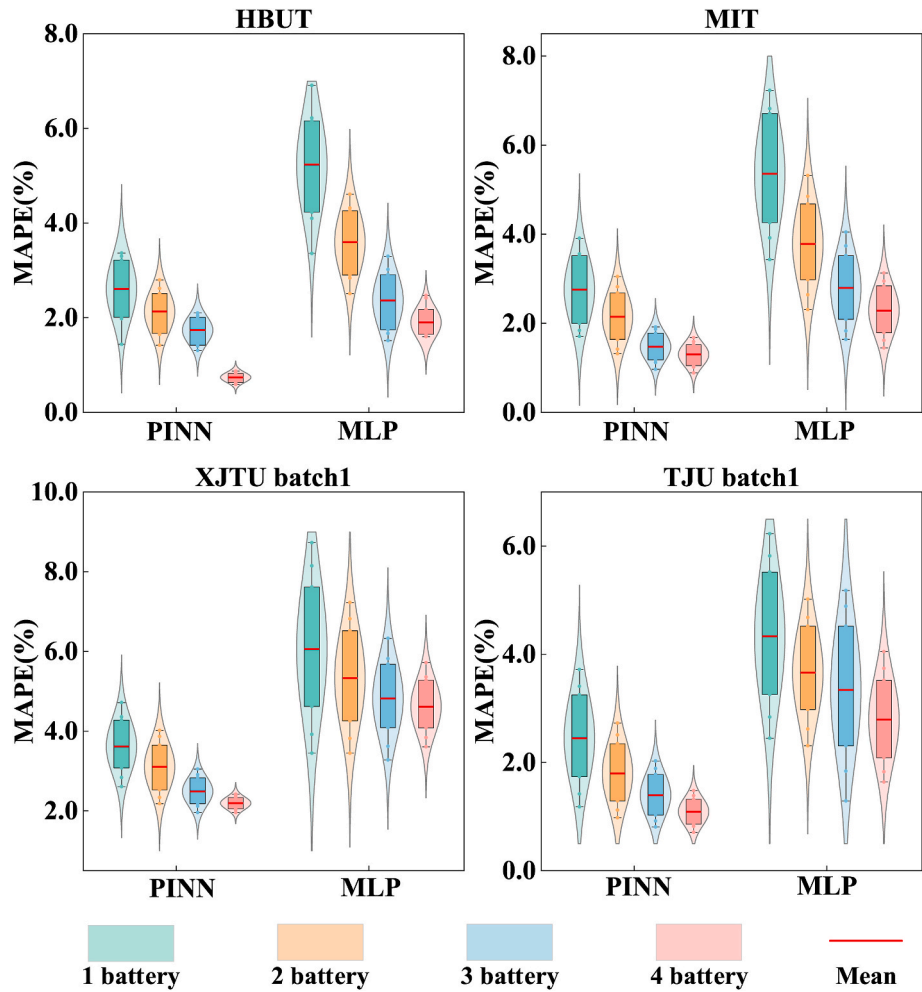


Fig. 5. shows the SOH estimation results of small sample experiments. The MAPE distributions of PINN and MLP models on four datasets from 1 battery to 4 battery training. Each error bar represents 10 independent experimental trials, with the central marker indicating the mean value.

Table 4
MAPE for fine-tuning experiments on four datasets. All values are expressed in percentage.

	Fine-tuning				Non-tuning			
	HBUT	MIT	XJTU	TJU	HBUT	MIT	XJTU	TJU
HBUT	–	1.17	2.40	3.69	–	9.47	11.38	15.60
MIT	0.91	–	1.47	3.91	9.19	–	14.11	17.76
XJTU	2.69	1.45	–	1.09	11.30	14.73	–	8.98
TJU	1.61	1.36	3.46	–	11.96	10.99	20.88	–

model with massive data in the source domain. The results show that the error fluctuation in the migration scenario is reduced by about 25 %. These findings confirm the unique value of physical information machine learning in battery management systems and provide an efficient development paradigm for a new generation of intelligent battery management systems by integrating mechanism knowledge and experimental data.

The PINN method combines the equations of the traditional physical model with the high-dimensional nonlinear fitting characteristics of the neural network, which improves the adaptability of the model while maintaining the physical interpretability. However, there are still many challenges in this field. On the one hand, it is impossible to compare the measured SEI film growth dynamics with the simulated SEI film volume growth. On the other hand, the existing methods cannot achieve the perfect fusion of mechanism model and neural network. Our proposed scheme achieves a good balance under the existing data and technical

conditions, which lays a foundation for subsequent research. With the advancement of testing technology and the availability of more internal state variables, the introduction of more comprehensive electrochemical models can be considered in the future.

CRediT authorship contribution statement

Aina Tian: Writing – review & editing, Validation, Methodology. **Luyao He:** Writing – original draft, Validation, Software, Methodology, Formal analysis, Data curation. **Tao Ding:** Software, Data curation. **Kailang Dong:** Methodology, Formal analysis. **Yuqin Wang:** Validation, Methodology. **Jiuchun Jiang:** Writing – review & editing, Methodology, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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